

Ex.no:4	EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION

Aim:

To perform data inspection, filtering, grouping, aggregation, descriptive statistical analysis, and correlation analysis using Pandas and visualize the results using Matplotlib.

Algorithm:

1. Import the required Pandas, NumPy and Matplotlib libraries.
2. Read the stud_data.csv file using Pandas.
3. Display the dataset and inspect the first few records.
4. Find the number of rows and columns and display the data types.
5. Generate the statistical summary of the dataset.
6. Filter the students based on marks and attendance conditions.
7. Sort the student records based on marks and attendance.
8. Group the data based on department and calculate average marks and attendance.
9. Calculate mean, median, minimum, maximum and standard deviation of marks.
10. Calculate the correlation between marks and attendance.
11. Create bar chart, line chart, scatter plot, histogram and box plot using Matplotlib.
12. Display the results and interpret the obtained data.

Program:

```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Read dataset
df = pd.read_csv("stud_data.csv")

# Display dataset
print(df)

# Display first five records
print(df.head())

# Display shape
print(df.shape)

# Display data types
print(df.dtypes)

# Display statistical summary
print(df.describe())

# Filter students with marks greater than 80
print(df[df["Marks"] > 80])

# Filter students with attendance greater than 90
print(df[df["Attendance"] > 90])

# Sort students by marks
print(df.sort_values("Marks", ascending=False))
```

```
# Sort students by attendance
print(df.sort_values("Attendance", ascending=False))

# Department-wise average marks
print(df.groupby("Department")["Marks"].mean())

# Department-wise average attendance
print(df.groupby("Department")["Attendance"].mean())

# Calculate mean
print("Mean Marks:", df["Marks"].mean())

# Calculate median
print("Median Marks:", df["Marks"].median())

# Calculate minimum
print("Minimum Marks:", df["Marks"].min())

# Calculate maximum
print("Maximum Marks:", df["Marks"].max())

# Calculate standard deviation
print("Standard Deviation:", df["Marks"].std())

# Display correlation matrix
print(df[["Marks", "Attendance"]].corr())

# Display correlation between marks and attendance
print("Correlation:", df["Marks"].corr(df["Attendance"]))

# Bar chart
```

```
plt.bar(df["Name"], df["Marks"])
plt.title("Student Marks")
plt.xlabel("Name")
plt.ylabel("Marks")
plt.show()
```

```
# Line chart
plt.plot(df["Name"], df["Attendance"], marker="o")
plt.title("Student Attendance")
plt.xlabel("Name")
plt.ylabel("Attendance")
plt.show()
```

```
# Scatter plot
plt.scatter(df["Attendance"], df["Marks"])
plt.title("Marks vs Attendance")
plt.xlabel("Attendance")
plt.ylabel("Marks")
plt.show()
```

```
# Histogram
plt.hist(df["Marks"], bins=5)
plt.title("Marks Distribution")
plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.show()
```

```
# Box plot
plt.boxplot(df["Marks"])
plt.title("Marks Box Plot")
plt.ylabel("Marks")
```

```
plt.show()
```

Output:

	Roll No	Name	Department	Marks	Attendance
0	101	Arun	CSE	85	92
1	102	Bala	IT	78	88
2	103	Divya	CSE	95	96
3	104	Priya	AI&DS	90	94
4	105	Ravi	IT	72	85
5	106	Karthik	AI&DS	88	91

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```
(6, 5)
```

```
Roll No      int64
```

```
Name        object
```

```
Department  object
```

```
Marks       int64
```

```
Attendance  int64
```

```
dtype: object
```

	Roll No	Marks	Attendance
count	6.000000	6.000000	6.00
mean	103.500000	84.666667	91.00
std	1.870829	8.382521	4.00
min	101.000000	72.000000	85.00
25%	102.250000	79.750000	88.75
50%	103.500000	86.500000	91.50
75%	104.750000	89.500000	93.50
max	106.000000	95.000000	96.00

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Department

AI&DS 89.0

CSE 90.0

IT 75.0

Name: Marks, dtype: float64

Department

AI&DS 92.5

CSE 94.0

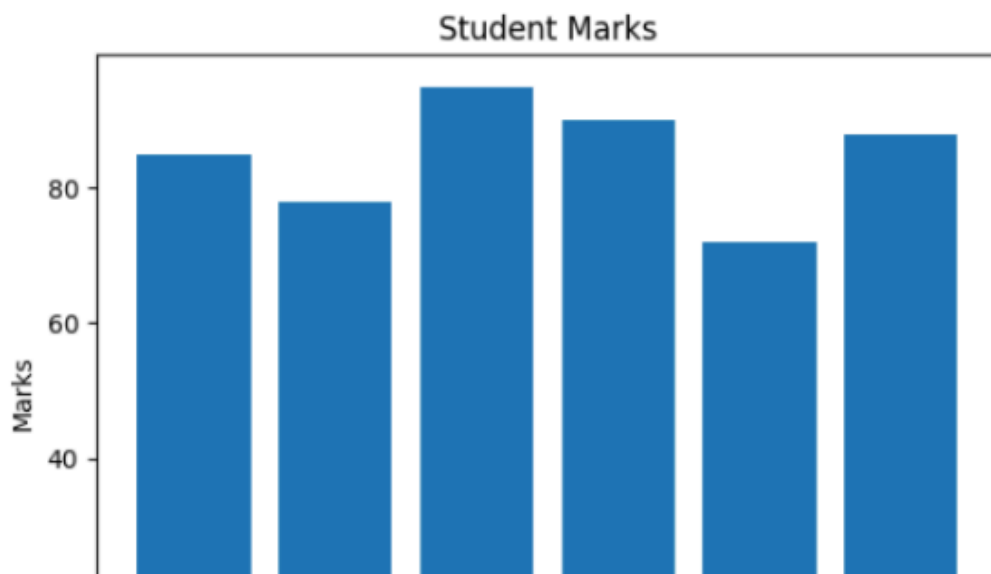
IT 86.5

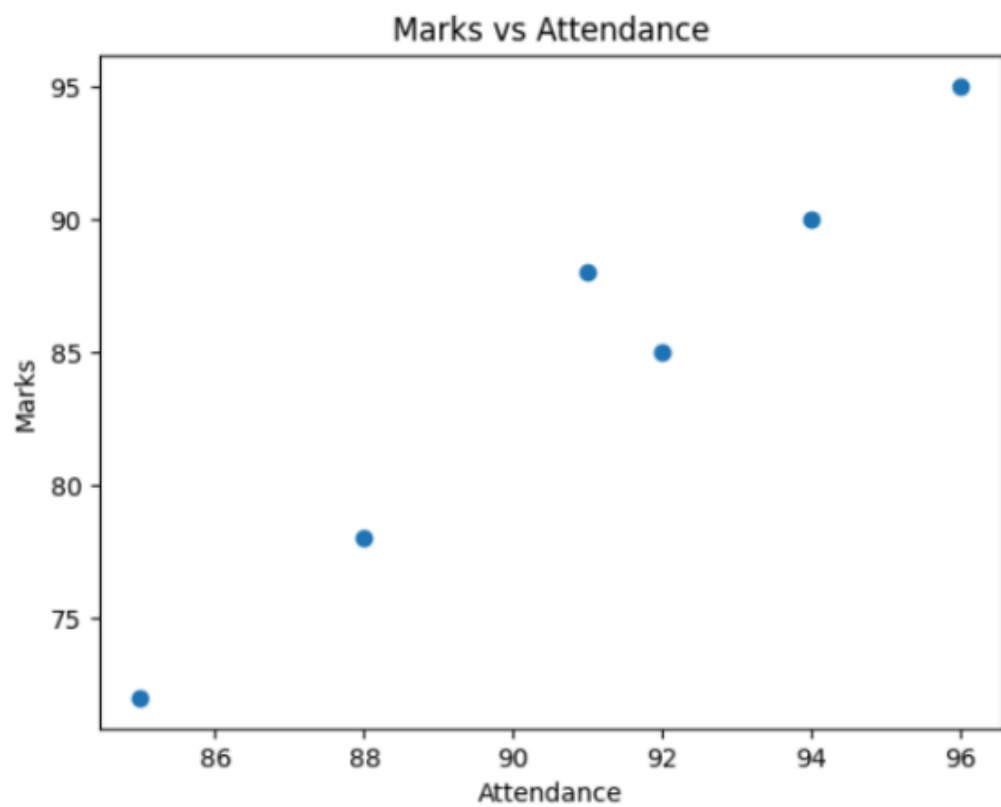
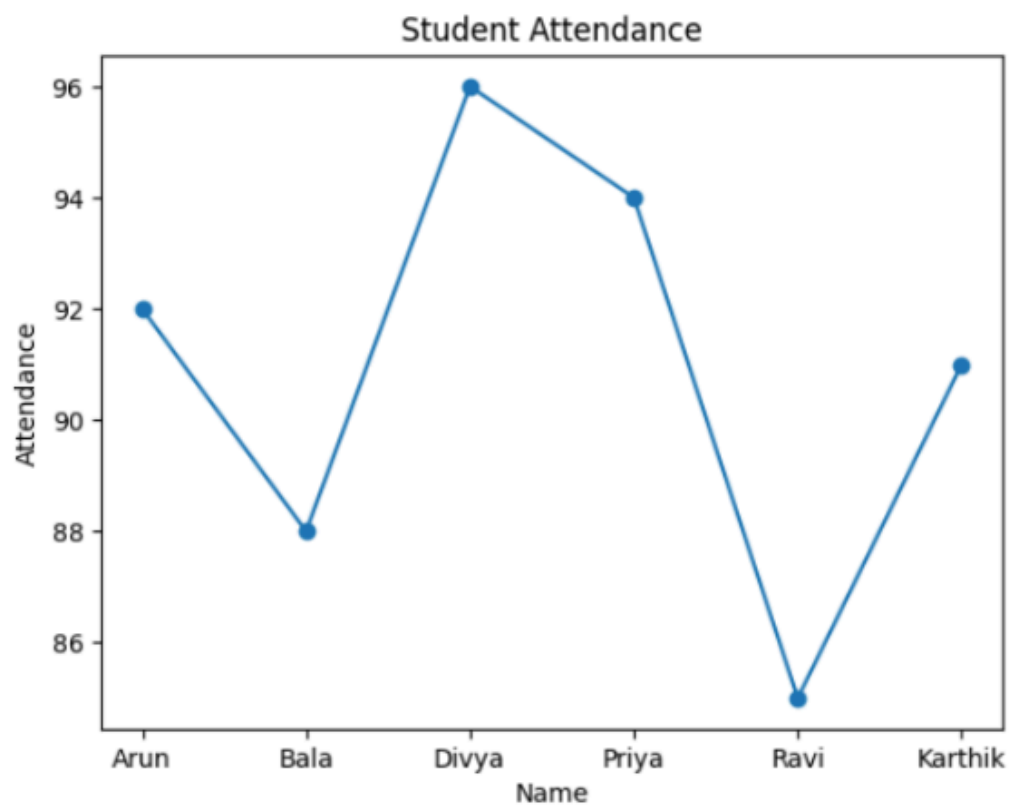
Name: Attendance, dtype: float64

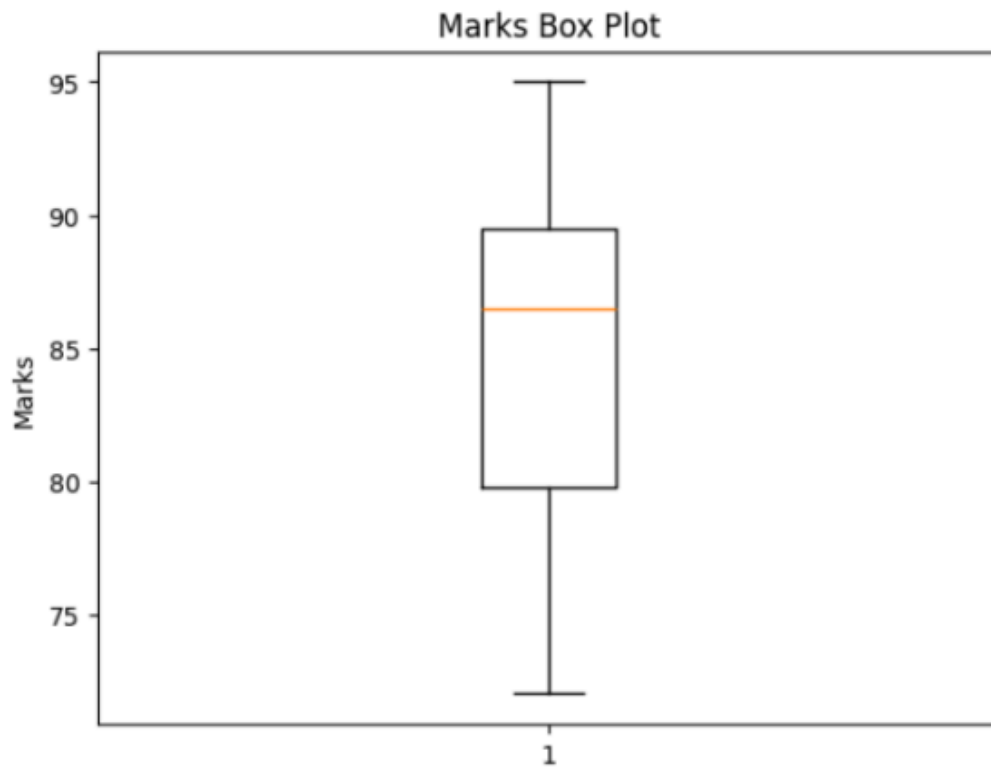
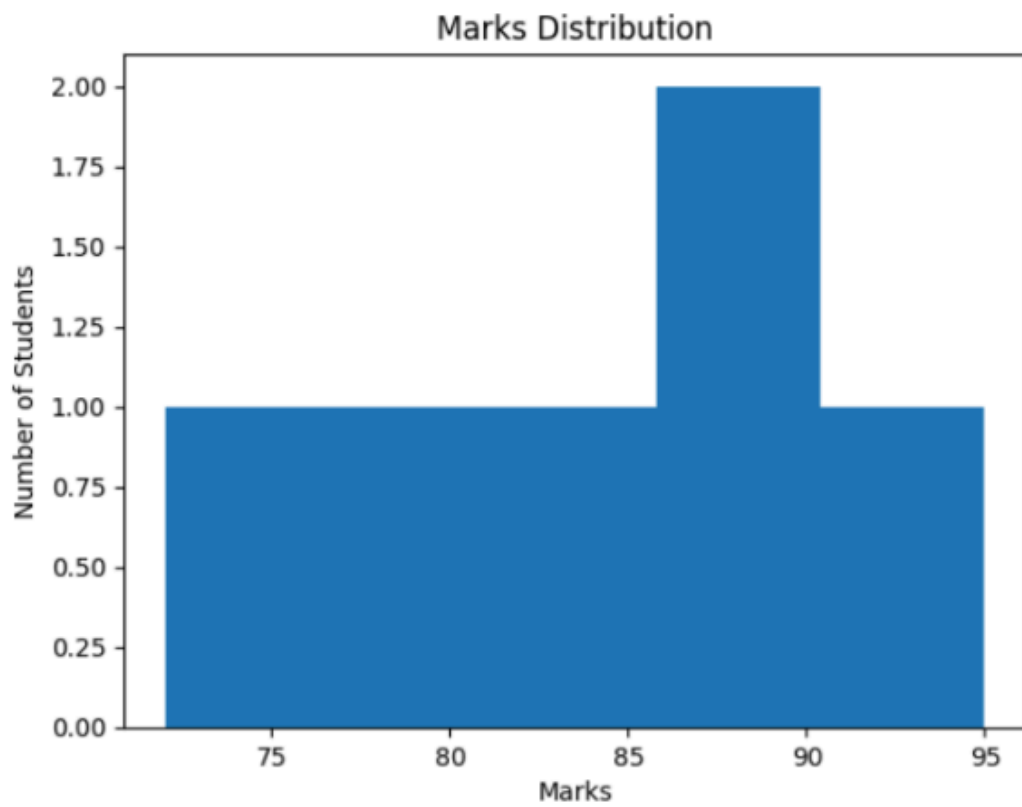
Mean Marks: 84.66666666666667
Median Marks: 86.5
Minimum Marks: 72
Maximum Marks: 95
Standard Deviation: 8.382521498133285

	Marks	Attendance
Marks	1.000000	0.978226
Attendance	0.978226	1.000000

Correlation: 0.9782259433305447







Result: